

Feature Ablation Study for Deep Learning Statistical Arbitrage on IDX Equities

trading-BEI project

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Abstract

We describe the experimental design for determining the optimal number and composition of input features for a deep learning statistical arbitrage (DLSA) strategy on Indonesia Stock Exchange (IDX) equities. Unlike the original DLSA pipeline, no factor model is used: normalized raw features are fed directly into a Transformer that learns representation and trading signal end-to-end. Because the deployed strategy is *long-only*, it carries market beta and is therefore benchmarked against a buy-and-hold IHSG index. We propose a nested, group-wise ablation across five economically motivated feature groups, run over eight random seeds per configuration, and evaluated on strictly time-ordered walk-forward out-of-sample data. The optimal feature set is identified as the point at which the walk-forward Sharpe ratio saturates before additional features degrade performance through overfitting.

1 Introduction

The **trading-BEI** project adapts Deep Learning Statistical Arbitrage (Guijarro-Ordóñez, Pelger & Zanotti, 2021) to IDX equities. The key deviation from upstream DLSA is the removal of the factor model and residual stages: raw daily stock-summary data is normalized and passed directly to a Transformer trading policy.

A central open question is how many input features the model should receive. Because there is no factor model to pre-extract structure, the Transformer must learn cross-asset co-movement itself. Adding features increases model capacity but, on a limited history of roughly 1,250 trading days, also increases the risk of overfitting. This report specifies a controlled ablation study to locate the optimal feature count and composition.

2 Data

The raw input is the IDX daily *Ringkasan Saham* (stock summary), one record per (date, ticker), spanning approximately five years (2020–2025). After cleaning, the panel retains 19 columns including OHLC prices, previous close, volume, value, trade frequency, foreign buy/sell,

Table 1: Feature groups used in the ablation.

Grp	Features	#
G1	log_return, mom_5/10/20	4
G2	hl_range, roll_vol_20, range_ratio	3
G3	log_volume, log_value, turnover, amihud	4
G4	foreign_flow_ratio, foreign_roll_5	2
G5	bid_ask_spread, book_imbalance	2

listed and tradeable shares, and end-of-day bid/offer.

Two data-quality issues are noted. First, the previously built `panel.parquet` is corrupt (unreadable footer) and must be regenerated. Second, the raw `OpenPrice` field is frequently zero; open-based features must fall back to the previous close or be flagged invalid to avoid dropping large numbers of rows.

3 Strategy Setting

The strategy is *long-only*: positions lie in $[0, 1]$ rather than the dollar-neutral $[-1, 1]$ of classical statistical arbitrage. The model head therefore uses a sigmoid (or clamped `tanh`), and weights are normalized to a fully-invested (or cash-permitting) book.

A direct consequence is exposure to market beta: unlike a market-neutral long/short book, a long-only strategy moves with the index. The appropriate benchmark is therefore a buy-and-hold IHSG index, not an absolute Sharpe threshold. We recommend training with the full cross-sectional signal (allowing the model to learn both out- and under-performers) and applying the long-only constraint only at the portfolio-construction stage.

4 Feature Groups

Ablation is performed over *groups* of economically related features rather than individual features, to limit the number of experiments and reduce noise. Each feature is standardized cross-sectionally within each trading day, following the DLSA convention. Table 1 lists the five groups.

Representative feature definitions, all causal (no look-

Table 2: Nested experiment design.

Exp	Active groups	$\approx \#$ feat.
A (baseline)	G1	4
B	G1+G2	7
C	G1+G2+G3	11
D	G1+G2+G3+G4	13
E (full)	G1+G2+G3+G4+G5	15

ahead):

$$r_t = \ln(C_t/C_{t-1}) \quad (1)$$

$$\text{hl_range}_t = (H_t - L_t)/C_t \quad (2)$$

$$\text{turnover}_t = V_t/\text{shares}_t \quad (3)$$

$$\text{amihud}_t = |r_t|/\text{value}_t \quad (4)$$

$$\text{ff_ratio}_t = (FB_t - FS_t)/(V_t + 1) \quad (5)$$

where C, H, L are close/high/low, V volume, FB/FS foreign buy/sell.

5 Ablation Design

Experiments are *nested*: starting from a minimal set and adding one group at a time (Table 2). The optimal feature count is the stage at which the walk-forward Sharpe stops improving.

If the Sharpe rises through A→B→C then plateaus or declines at D/E, the saturation point identifies the optimal feature set. An optional leave-one-group-out pass on the best configuration confirms each group’s marginal contribution.

6 Experimental Controls

For the comparison to be valid, *only the active feature list* may differ between experiments. All of the following are held identical: Transformer architecture (d_{model} , layers, heads, dropout); lookback T and target horizon; walk-forward split dates; transaction cost (bps), turnover penalty, and long-only constraint; epochs, learning rate, and early-stopping rule; and the seed grid. Without these controls, differences in Sharpe cannot be attributed to features.

7 Evaluation

Evaluation is out-of-sample (walk-forward), never training loss. The primary ranking metric is the annualized Sharpe ratio; supporting metrics are cumulative return versus buy-and-hold IHSG (because long-only carries beta), maximum drawdown, and turnover (a proxy for the high real trading costs on IDX).

For robustness, each configuration is run over **eight random seeds**, reporting mean $\pm$ standard deviation.

Eight seeds provide a stable variance estimate and support significance testing between configurations; a Sharpe gap below roughly 0.1 is likely seed noise rather than a feature effect. The full study therefore comprises $5 \times 8 = 40$ runs.

8 Implementation Changes

The study requires: (i) making `src/preprocess.py` config-driven, replacing the fixed `FEATURE_COLUMNS` constant with group definitions and computing the feature superset while selecting active groups from config; (ii) fixing the `open=0` bug for open-based features; (iii) one YAML per experiment in `configs/` differing only in `feature_groups`; (iv) extending `compare.py` to aggregate results into a single table (feature set $\rightarrow$ Sharpe $\rightarrow$ return $\rightarrow$ drawdown $\rightarrow$ turnover, mean $\pm$ std across seeds); and (v) saving metrics and equity-curve plots to `results/`.

9 Risks

The main risks are overfitting from limited data combined with many features (precisely the effect under test), feature collinearity (e.g. `log_volume` vs. `log_value` should not be read as new information), seed noise (hence multi-seed runs), and look-ahead leakage in momentum, rolling, and normalization steps (the most bug-prone area, to be covered by anti-look-ahead unit tests). The long-only benchmark must remain IHSG, not an absolute return figure.

10 Definition of Done

The study is complete when all five experiments (A–E) have run over eight seeds each (40 runs), the comparison table is populated with mean $\pm$ std for all metrics, the optimal feature set is identified with justification (the Sharpe saturation point), and findings are documented in `results/` and summarized in the project plan.